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BRANCH AND SECTION:CSM-B

COURSE NAME: FULL STACK WEB DEVELOPMENT

COURSE CODE:23CM4121

Experiment:04

Date:

Title: Develop Dynamic Webpage Using Java Script

Objective:

To create a dynamic weather dashboard that uses geolocation and API requests to display the user's city and current temperature with a responsive background color.

System Requirements:

- Hardware: Processor (e.g., Core i5), RAM (e.g., 8GB), Storage.
- Software: Operating System (e.g., Windows 11 / Linux), Code Editor (e.g., VS Code), Browser (e.g., Chrome), Runtime/Database (e.g., Node.js, MongoDB). [1, 2]

Source Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Weather DashBoard</title>
</head>
<body id="body">
  <form id="weatherForm">
```

```

<input type="text" id="city" placeholder="Enter City">
<button type="submit">Search</button>
<button type="button" id="getLocation">Get Current Location</button>
<button type="button" id="getTemp">Get Current temprature</button>
</form>
<h2 id="SelectedCity"></h2>
<h2 id="temperature"></h2>
</body>
<script>
  document.getElementById("weatherForm").addEventListener("submit", function(e) {
    e.preventDefault(); // Prevent page refresh

    const city = document.getElementById("city").value;
    document.getElementById("SelectedCity").innerText = city;
    // getWeather(city);
  });

  document.getElementById("getLocation").addEventListener("click", function() {
    if (navigator.geolocation) {
      navigator.geolocation.getCurrentPosition(async function(position) {
        const lat = position.coords.latitude;
        const lon = position.coords.longitude;
        // getWeatherByCoordinates(lat, lon);
        try{
          const response = await fetch(
            `https://nominatim.openstreetmap.org/reverse?format=json&lat=${lat}&lon=${lon}`
          );
          const data = await response.json();
          const name = document.getElementById("SelectedCity");
          name.innerText = data.address.city || data.address.town || data.address.village ||
            "Unknown Location";
        }
      });
    }
  });

```

```

    }

    catch (error) {

        console.error("Error fetching location data:", error);

    }

});

}

});

document.getElementById("getTemp").addEventListener("click", function() {

    if (navigator.geolocation) {

        navigator.geolocation.getCurrentPosition(async function(position) {

            const lat = position.coords.latitude;

            const lon = position.coords.longitude;

            try {

                const response = await fetch(

                    `https://api.openweathermap.org/data/2.5/weather?lat=${lat}&lon=${lon}&appid=280668dd93f8ac56029ab0ecf13967b1`

                );

                const data = await response.json();

                // Process temperature data

                console.log("Data:",data)

                document.getElementById("temperature").innerText = `Temperature: ${data.main.temp - 273.15} °C`;

                if (data.main.temp - 273.15 > 30) {

                    document.getElementById("body").style.backgroundColor = "red";

                }

                else{

                    document.getElementById("body").style.backgroundColor = "blue";

                }

            } catch (error) {

                console.error("Error fetching temperature data:", error);

            }

        });

    }

});

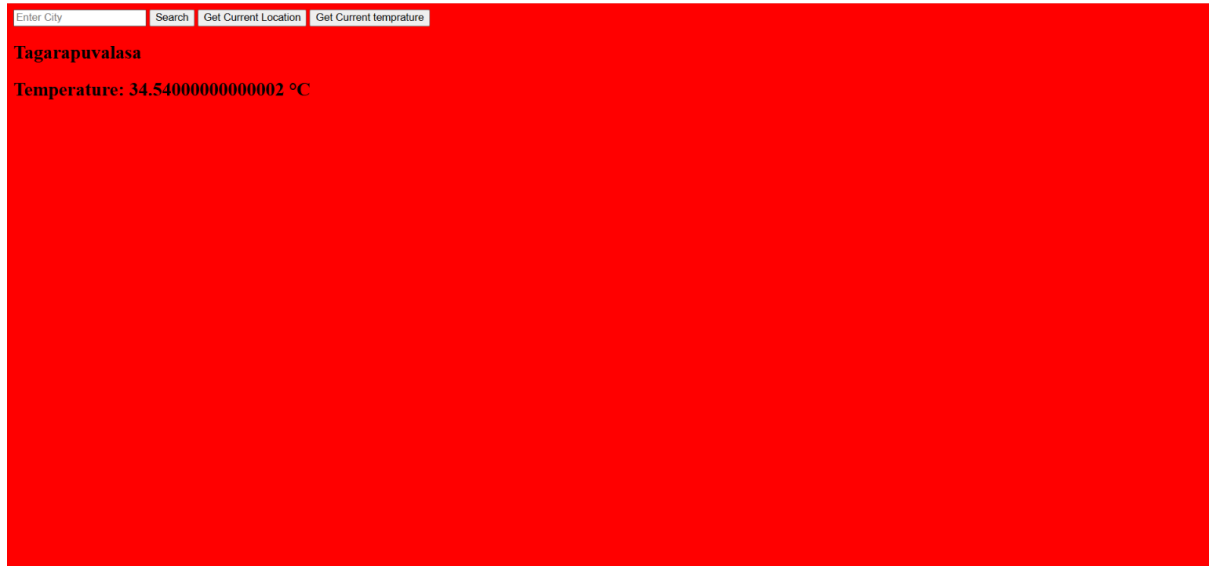
```

```
    });  
  }  
});  
</script>  
</html>
```

STEPS TO EXECUTE:

- 1) Open vs code
- 2) Click on run debugging
- 3) Click on Browser
- 4) Click on Chrome and run

OUTPUT:



Result: Successfully executed a dynamic webpage using HTML, CSS, JS